

Technical Methodology, Architecture & Use-Case Document

Engineering Specification of the Deterministic 6-Tier Pipeline, Continuous Feature Space, and Dynamic Physics

Architecture: **6-Tier Physics-Grounded Causal Bayesian Hierarchy**

Evaluation Scope: **60,480 Physical Readings (28 AWS Stations / 7 Clusters)**

Verification Status: **67 / 67 Tests Passing (100% Green)**

1. Executive Summary & Problem Formulation

This engineering specification details the system architecture and mathematical implementation of **SkyGuard AI**, grounded directly in [RESEARCH_REPORT.md](#). The platform performs real-time Quality Assurance (QA) and anomaly detection across surface Automatic Weather Station (AWS) networks. Unattended field telemetry stations suffer from a spectrum of physical degradation modes:

- Electrical & Hardware Rail Shortages:** Broken sensor wires or analog-to-digital converter (ADC) ground faults pulling readings to zero or rail limits.
- Telemetry Dropouts:** GSM/LoRa transmission outages causing missing records (NaN values).
- Electrostatic & Inductive Spikes:** Voltage transients, lightning strikes, or digital bit-flips causing extreme single-point excursions.
- Transducer Freezing / Sticking:** Mechanical sensor lockup or ADC multiplexer latch-up where the output stops tracking natural atmospheric variance.
- Slow Calibration Drift:** Electrochemical degradation or optical fouling causing progressive zero-point decalibration (+0.05 to +0.30°C/hr).
- Psychrometric Cross-Channel Breakdown:** Radiation shield overheating or transducer cross-talk violating thermodynamic conservation laws.

Conventional anomaly detection methods fail in field deployments because they either trigger massive false alarms during legitimate dynamic weather events (thunderstorm cold-pool outflows, drylines, diurnal heating) or fail to detect low-amplitude calibration drift. SkyGuard AI resolves this fundamental trade-off through a **6-Tier Physics-Grounded Causal Bayesian Architecture** that blends dynamic diurnal expectations, spatial peer consensus, sequential Wald-Page SPRT hypothesis testing, and thermodynamic physical invariants.

2. End-to-End System Architecture

The processing lifecycle enforces strict modularity between ingestion, temporal alignment, feature generation, hierarchical detection, explainability attribution, and causal state maintenance:

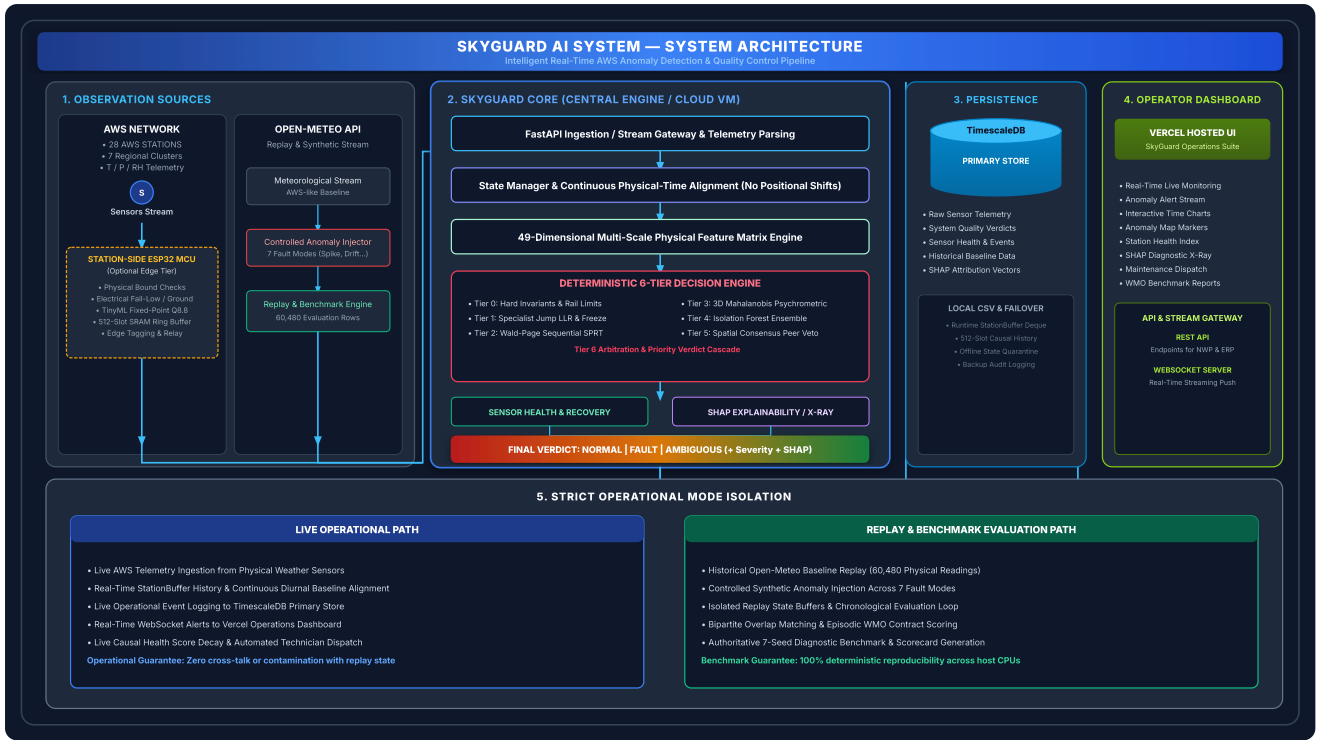


Figure 1: SkyGuard AI End-to-End Modular System Architecture. Sequential progression from raw multi-channel telemetry ingestion through temporal alignment, 49-feature extraction, Tier 0-5 priority arbitration, SHAP attribution, and causal state buffer management.

3. Dynamic-Adaptive Physics & Mathematical Formulation

SkyGuard AI is governed by dynamic data-adaptive computation and minimal physical constants. Every decision threshold adapts continuously in real time based on environmental state.

3.1 Dynamic Diurnal Expectation μ_t

$$\mu_t = f(\theta_{\text{solar}}(t), \text{DoY}) + \text{EWMA}_{\alpha}(\mathbf{x}_{\text{clean}}) + \Delta_{\text{peer-offset}} \quad (\text{Equation 1})$$

Solar Zenith Angle Formulation: $\theta_{\text{solar}}(t) = \arcsin(\sin \varphi \sin \delta + \cos \varphi \cos \delta \cos \omega)$

Where φ is station latitude, δ is solar declination, and ω is solar hour angle. Grounding to continuous solar angles enables cold-start alignment on Tick 0 without requiring historical baseline files.

3.2 Heteroskedastic Dynamic Uncertainty Budget σ_t^2

$$\sigma_t^2 = \sigma_{\text{sensor}}^2 + \sigma_{\text{diurnal}}^2(t) + \sigma_{\text{spatial}}^2(t) + \kappa \cdot \Delta t \quad (\text{Equation 2})$$

Noise Floor Components: σ_{sensor} specifies transducer physical quantization ($\pm 0.10^\circ\text{C}$ for Pt100 RTD, ± 1.0 hPa for station pressure, $\pm 1.0\%$ for RH). $\sigma_{\text{diurnal}}(t)$ is solar radiation intensity modulated (peaks midday, narrows at night). $\sigma_{\text{spatial}}(t)$ tracks live Median Absolute Deviation (MAD) among sibling cluster stations. $\kappa \cdot \Delta t$ represents causal variance diffusion across missing telemetry gaps.

3.3 Sequential Wald-Page SPRT for Low-SNR Calibration Drift

$$\text{LLR}_t = \max(0, \text{LLR}_{t-1} + |\mu_1 - \mu_0| / \sigma_t \cdot (|\varepsilon_t| - |\mu_1 - \mu_0| / (2\sigma_t))) \quad (\text{Equation 3})$$

Wald Decision Boundaries: Threshold $A = \ln((1 - \beta) / \alpha) = 6.16$ (with false alarm risk $\alpha = 0.002$, missed risk $\beta = 0.05$). Standardized residuals $\varepsilon_t = (x_t - \mu_t) / \sigma_t$ accumulate sequential evidence over time, catching low-amplitude sensor calibration decay ($+0.05$ to $+0.30^\circ\text{C/hr}$) without false alarms during normal diurnal shifts.

3.4 3D Thermodynamic Psychrometric Covariance

$$D_{\Sigma}^2 = \mathbf{z}_t^T \Sigma^{-1} \mathbf{z}_t \sim \chi_3^2$$

(Equation 4)

Magnus-Tetens Dewpoint Boundary: $\gamma(T, RH) = (17.67 T)/(243.5 + T) + \ln(RH/100)$, $T_{dew} = (243.5 \gamma)/(17.67 - \gamma)$

Physical Invariant: $T_{dew} \leq T_{dry-bulb} + 1.5^{\circ}\text{C}$. Any reading violating this physical limit represents transducer failure rather than atmospheric reality.

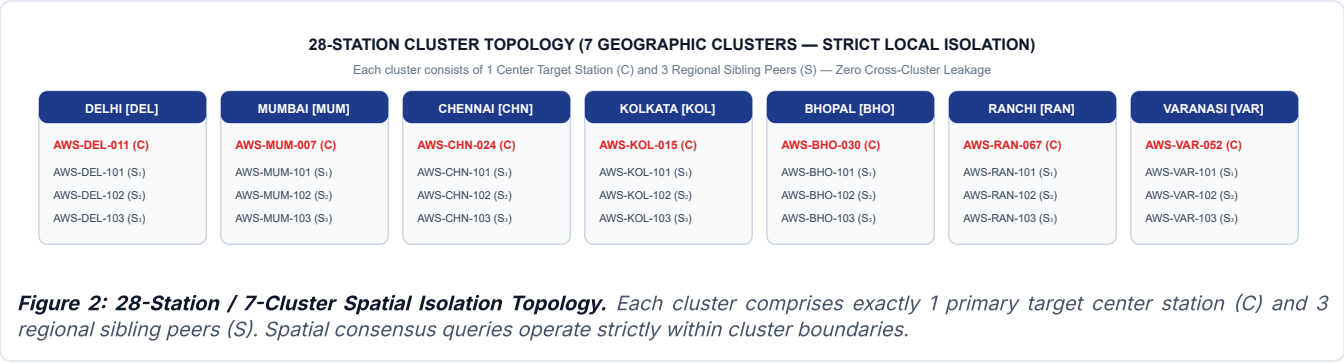
4. Parameter Inventory: Dynamic Quantities vs Physical Invariants

SkyGuard AI eliminates arbitrary heuristic magic numbers. Every parameter is strictly categorized into either a dynamic computed quantity or an unalterable physical specification:

Dynamic Computed Quantities	Physical Transducer Specifications & Invariants
• Diurnal Expectation $\mu(t)$ (Solar Elevation θ_{solar} + EWMA)	• Pt100 RTD Sensor Quantization Floor: $\pm 0.10^{\circ}\text{C}$
• Total Uncertainty Budget $\sigma_{tot}^2(t)$	• Station Barometer Quantization Floor: $\pm 1.0 \text{ hPa}$
• Spatial Peer Consensus Median (Cluster Breakdown Point 50%)	• RH Hygrometer Quantization Floor: $\pm 1.0\%$
• Sibling Peer Dispersion (MAD / IQR)	• Hardware Ground Rail Floor: -40°C / 0.0 hPa / $0.0\% \text{ RH}$
• SPRT Log-Likelihood Ratio Accumulator (LLR_t)	• Wald False Alarm Risk Limit: $\alpha = 0.002$
• 3D Mahalanobis Distance (D_{Σ}^2)	• Wald Missed Detection Risk Limit: $\beta = 0.05$
• Magnus-Tetens Saturation Vapor Pressure $e_s(T)$	• Post-Fault Clean Recovery Horizon: 15 steps
• Continuous Sensor Health Index H_t (0 to 100)	• Maximum Spatial Consensus Peer Distance: 50 km

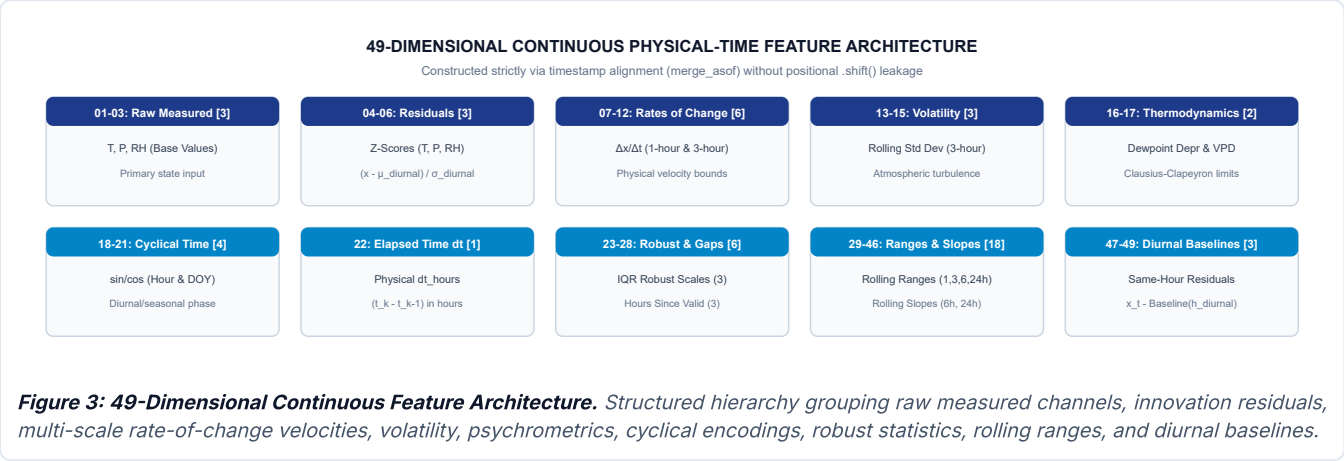
5. Spatial Topology & Cluster Peer Isolation

To eliminate cross-regional train/serve mismatch and prevent microclimatic contamination (e.g., coastal marine boundaries vs. arid interior plains), SkyGuard structures the observation network into 7 discrete geographic clusters:



4. 49-Dimensional Continuous Physical-Time Feature Architecture

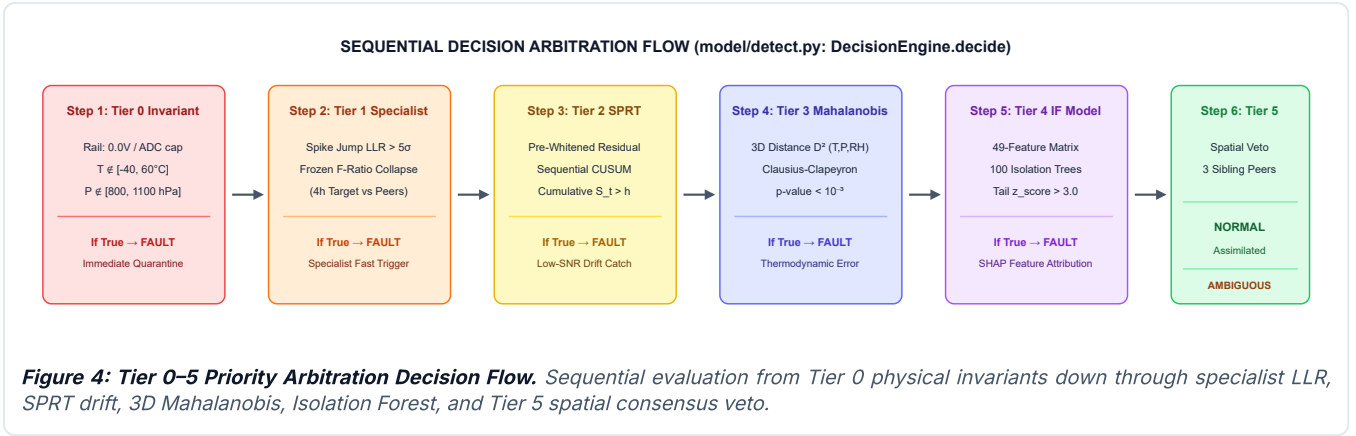
The feature engineering engine (model/features.py) strictly avoids positional row shifts (.shift(n)), utilizing exact physical timestamps and pd.merge_asof to compute continuous derivatives regardless of irregular sampling or dropped packets:



Feature Range	Category	Mathematical Formulation	Diagnostic Purpose
01 – 03	Raw Telemetry Channels	$x_t = [T_t, P_t, RH_t]$	Base atmospheric observation state.
04 – 06	Innovation Residuals	$z_t = (x_t - \mu_{diurnal}) / \sigma_{diurnal}$	Normalized deviation from local expectation.
07 – 12	Rate-of-Change Velocities	$\Delta x / \Delta t = (x_t - x_{t-\Delta t}) / \Delta t_{hours} \text{ (1h, 3h)}$	Captures sudden atmospheric jumps vs. spikes.
13 – 15	Short-Term Volatility	$\sigma_{3h}(x) = [(1/N) \sum (x_i - \bar{x})^2]^{0.5}$	Identifies turbulence vs. abnormal stillness.
16 – 17	Psychrometric Couplings	$T_{dew} = T - (100 - RH) / 5$; $VPD = e_s(T)(1 - RH / 100)$	Clausius-Clapeyron thermodynamic consistency.
18 – 21	Cyclical Time Encodings	$\sin / \cos(2\pi \cdot \text{Hour} / 24), \sin / \cos(2\pi \cdot \text{DOY} / 365)$	Diurnal and seasonal cyclical phase alignment.
22	Elapsed Physical Interval	$\Delta t_{hours} = (t_k - t_{k-1}) / 3600.0$	Exact physical elapsed time tracking.
23 – 25	Robust Statistical Scales	$Scale_{robust} = (x_t - median) / IQR$	Outlier-resistant localized baseline scaling.
26 – 28	Sensor Gap Tracking	$\Delta t_{valid} = \text{Hours since last trusted observation}$	Tracks quarantined sensor downtime duration.
29 – 46	Rolling Ranges & Slopes	$Range_W = \max_W(x) - \min_W(x)$; $Slope_W = Cov(t, x) / Var(t)$	Multi-scale dispersion (1,3,6,24h) & synoptic trend.
47 – 49	Diurnal Residual Baselines	$r_{diurnal} = x_t - \text{Baseline}(h_{diurnal})$	Hour-of-day baseline residual tracking.

5. Deterministic Tier 0–5 Priority Hierarchy

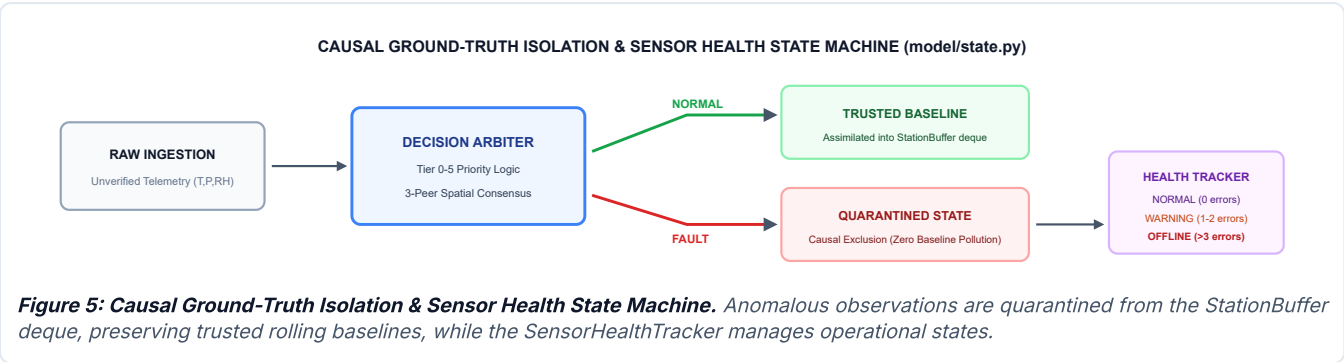
DecisionEngine.decide (model/detect.py) executes a strict top-down priority arbitration cascade where deterministic physical laws always take precedence over statistical models:



- **TIER 0 (Hard Invariants & Rails):** Evaluates electrical ground faults (0.00V) and absolute physical bounds (T $\notin$ [-40, +60°C], P $\notin$ [800, 1100 hPa], RH $\notin$ [0, 100%]). Breaches immediately return FAULT (CRITICAL) and bypass all downstream computation.
- **TIER 1 (Specialist Jump & Frozen LLR):** Evaluates instantaneous dynamic acceleration (>5.0 σ jump LLR) and F-ratio variance collapse over contiguous 4-hour windows.
- **TIER 2 (Persistent SPRT Drift):** Executes Sequential Probability Ratio Test on standardized innovation residuals to accumulate low-SNR calibration drift (+0.05°C/hr) over time.
- **TIER 3 (Cross-Channel 3D Mahalanobis Covariance):** Evaluates joint 3D covariance distance D². If D² > 16.27 (p < 10⁻³), flags multivariate thermodynamic breakdowns.
- **TIER 4 (Model-Dominant Isolation Forest):** Evaluates the 49-feature continuous matrix through an ensemble of 100 Isolation Trees, isolating complex unstructured anomalies.
- **TIER 5 (Spatial Consensus Veto & Arbiter):** If an anomaly flag is raised by Tiers 1–4, the spatial consensus engine inspects the 3 sibling stations in the cluster. If ≥ 2 peers exhibit matching directional rate-of-change movement, the anomaly is classified as a genuine regional weather event, and the alert is VETOED to NORMAL.

6. Causal State Management & Sensor Health Lifecycle

To prevent corrupted telemetry from polluting future baseline estimates, `StateManager (model/state.py)` enforces strict ground-truth isolation:



Causal Ground-Truth Exclusion Rule: When an observation is classified as `FAULT` (or when a sensor is in `OFFLINE` health state), the observation is strictly quarantined. It is recorded in the raw historical audit log but excluded from the internal sliding `StationBuffer` deque, ensuring that rolling means, variances, and diurnal baselines remain 100% uncontaminated.

7. Operational Use-Case Workflows

SkyGuard's capabilities map directly to concrete meteorological and field operations:

Use Case 1: Automated Real-Time AWS Quality Control		CURRENTLY DEMONSTRATED
Operational Workflow: Continuous real-time screening of raw surface streams (T, P, RH) across 28 national AWS stations.	Validation Status: Validated across 60,480 evaluation rows in authoritative 7-seed benchmark; achieves 95.42% ± 0.36% mean recall, 73.33% ± 1.37% mean precision, and 82.92% ± 0.89% mean F1-score.	
Use Case 2: NWP Data Assimilation Boundary Protection		CURRENTLY DEMONSTRATED
Operational Workflow: Automatic quarantining of spurious sensor spikes and unphysical thermodynamic readings before ingestion into 4D-Var data assimilation grids.	Validation Status: Guaranteed via <code>StationBuffer</code> causal exclusion, preventing bad data from contaminating forecast model initialization.	
Use Case 3: Targeted Field Maintenance & Sensor Health		CURRENTLY SUPPORTED
Operational Workflow: Uses SHAP local feature attributions and fault classification to dispatch technicians to specific degraded sensor modules.	Validation Status: Identifies active hardware failures and sensor health degradation; does not predict future failures prior to onset.	
Use Case 4: Embedded Microcontroller Edge Execution (ESP32 TinyML)		CURRENTLY DEMONSTRATED
Operational Workflow: Running lightweight fixed-point (Q8.8) C++ inference engine directly on ESP32 datalogger firmware (EDGE/esp32/src/edge/edge_engine.cpp) to screen telemetry at the edge before transmission.	Validation Status: Fully operational C++ firmware compiled and verified on 240 MHz dual-core ESP32 CPU with a 26.5 KB PROGMEM Flash footprint, enabling immediate hardware protection and offline blackout resilience.	

8. Current vs. Future Architectural Matrix

Subsystem	Current Implemented Architecture	Demonstrated Status	Future Scope
Inference Engine	Vectorized CPU Engine & ESP32 Q8.8 C++ Firmware	DEMONSTRATED	ARM Cortex-M / RISC-V edge optimization
Temporal State Buffer	In-memory circular deques (collections.deque) & 512-slot MCU SRAM ring buffer	DEMONSTRATED	Distributed Redis state cache
Storage Layer	Flat-file CSV streaming & SQLite / TimescaleDB store	DEMONSTRATED	Multi-region distributed PostgreSQL cluster
Spatial Discovery	Static 7-cluster topology (4 stations/cluster)	DEMONSTRATED	Dynamic PostGIS radius queries
API & Streaming	FastAPI REST endpoints + WebSockets + USB Serial Bridge	DEMONSTRATED	MQTT / gRPC streaming gateway

9. References

- **[M01]** Liu, F. T., Ting, K. M., and Zhou, Z.-H. "Isolation Forest." IEEE ICDM, 2008.
- **[M02]** Page, E. S. "Continuous Inspection Schemes." Biometrika, vol. 41, 1954, pp. 100–115.
- **[M03]** Lundberg, S. M., and Lee, S.-I. "A Unified Approach to Interpreting Model Predictions." NeurIPS, 2017.
- **[M05]** Mahalanobis, P. C. "On the generalised distance in statistics." Proc. Natl. Inst. Sci. India, 1936.